

Exercise Sheet 4

Propositional Logic – Classical Reasoning

1. Provide an constructive Natural Deduction proof of $(A \vee \neg A) \rightarrow (\neg\neg A \rightarrow A)$.
2. Provide constructive Natural Deduction proof that $((P \rightarrow \perp) \rightarrow P) \rightarrow P$ (this is an instance of what is known as Peirce's law) implies $\neg\neg P \rightarrow P$, and vice versa.
3. Provide a classical Natural Deduction proof of $((P \rightarrow Q) \rightarrow P) \rightarrow P$.
4. Provide a classical Natural Deduction proof of $\neg(A \wedge B) \rightarrow (\neg A \vee \neg B)$.